Using Moments with Equilibrium

Exam code: 9MA0

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Using moments with equilibrium

What is the resultant moment for a body in equilibrium?

- If a rigid body is in **equilibrium** then:
  - The **resultant force** is 0 N
  - The **resultant moment** about **any point** is 0 N m
- The sum of the magnitudes of the **clockwise moments equals** the sum of the magnitudes of the **anti-clockwise moments**

How do I use F = ma and moments to solve equilibrium problems?

- Step 1: Identify **all forces** and **perpendicular distances** from the pivot
- Step 2: Use the fact that the **resultant force is 0 N** to form an equation
- Step 3: Use the fact that the **resultant moment** about any point is **0 N m** to form equations
  - If there is an unknown force you can choose a pivot that is on the line of action of that force – this will mean that the moment of that force is 0 N m

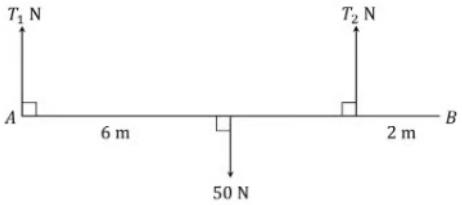
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

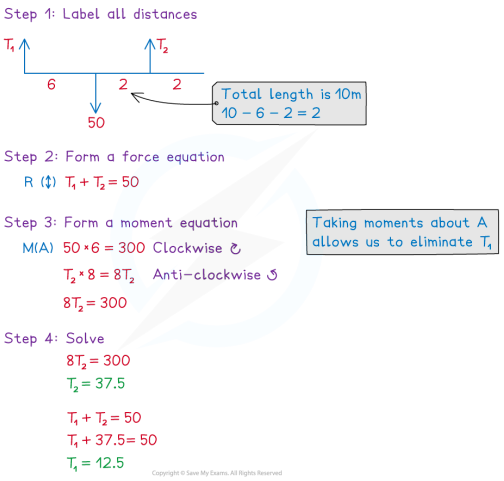
A light rod AB of length 10 m is held in equilibrium by three forces: 50 N, T_1 N and T_2 N.

T_1 N acts vertically upwards at A, 50 N acts vertically downwards 6 m away from A and T_2 N acts vertically upwards 2 m away from B as shown in the diagram.



Find the values of T_1 and T_2.

Answer:



Examiner Tips and Tricks

- If you see a force diagram that involves distances then you can be fairly sure that moments will be involved! A clear diagram showing all the distances is key.
- If you pick a pivot and form a moment equation which involves more than one unknown, try choosing another pivot to form simultaneous equations.
- Pick your pivot wisely as unknown and unwanted forces can be eliminated from the equation by choosing the point to be on the line of action of a force.

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